SHEN ZHENG

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4804 Forbes Ave, Pittsburgh, PA 15213

EDUCATION

Carnegie Mellon University, U.S. *2024.8 – present*
Ph.D. in Robotics, Robotics Institute Advisor: Prof. Srinivasa Narasimhan

Carnegie Mellon University, U.S. *2022.8 – 2023.12*
M.S. in Computer Vision, Robotics Institute Advisor: Prof. Srinivasa Narasimhan

Wenzhou-Kean University, China *2017.9 – 2021.6*
B.S. in Mathematical Sciences Advisor: Prof. Gaurav Gupta

RESEARCH INTEREST

Computer Vision, Generative and World Models, Efficiency, Autonomous Driving, Embodied AI

SELECTED PUBLICATIONS

MapLightning: Online Vectorized HD Map Construction with 1D Map Tokens
Shen Zheng, Anurag Ghosh, Mani Ramanagopal, Srinivasa Narasimhan
Under Review [arXiv] [Page]

WarpI2I: Image Warping for Image-to-Image Translation
Shen Zheng, Anurag Ghosh, Gaurav Parmar, Srinivasa Narasimhan
ECCV 2026 [arXiv] [Page]

ROADWork: A Dataset and Benchmark for Learning to Recognize, Observe, Analyze and Drive Through Work Zones
Anurag Ghosh, **Shen Zheng**, Robert Tamburo, Juan R. Alvarez Padilla, Hailiang Zhu, Michael Cardei, Nicholas Dunn, Christoph Mertz, Srinivasa Narasimhan
ICCV 2025 [arXiv] [Page]

Instance-Warp: Saliency Guided Image Warping for Unsupervised Domain Adaptation
Shen Zheng*, Anurag Ghosh*, Srinivasa Narasimhan
WACV 2025 [arXiv] [Page]

TPSeNCE: Towards Artifact-Free Realistic Rain Generation for Deraining and Object Detection in Rain
Shen Zheng, Changjie Lu, Srinivasa Narasimhan
WACV 2024 [arXiv] [Page]

Low-Light Image Enhancement: A Comprehensive Survey and Beyond
Shen Zheng, Yiling Ma, Jinqian Pan, Changjie Lu, Gaurav Gupta
Survey [arXiv]

PointNorm: Dual Normalization is All You Need for Point Cloud Analysis
Shen Zheng, Jinqian Pan, Changjie Lu, Gaurav Gupta
IJCNN 2023 Oral [arXiv] [Page]

Semantic-Guided Zero-Shot Learning for Low-Light Image/Video Enhancement
Shen Zheng, Gaurav Gupta
WACV 2022 [arXiv] [Page]

AS-IntroVAE: Adversarial Similarity Distance Makes Robust IntroVAE
Changjie Lu, **Shen Zheng**, Zirui Wang, Omar Dib, Gaurav Gupta
ACML 2022 [arXiv]

INDUSTRY EXPERIENCE

Waymo *Summer 2026*
Perception Software Engineer Intern, 3D Occupancy and World Model Mentor: Zhenyao Zhu

Waymo *Summer 2025*
Perception Software Engineer Intern, Online HD Map Construction Mentor: Guohao Zhang

Lucid Motors *2024.2 – 2024.7*
Perception Software Engineer, Traffic Light Detection Director: Dr. Feng Guo

Momenta *2021.9 – 2022.2*
Perception Engineer Intern, Traffic Light Detection Director: Dr. Wangjiang Zhu

ACADEMIC SERVICE

Conference Reviewer: CVPR, ICCV, ECCV, WACV, BMVC, AAAI, IJCNN
Journal Reviewer: TNNLS, IJCV, TCSVT, ESWA, EAAI, JVCIR, Neurocomputing